

COURSE RECOVERY SYSTEM (CRS)
OBJECT ORIENTED DEVELOPMENT
WITH JAVA (CT083-3-2-00DJ)



Java

INTAKE- APD2F2509CS(DA)
LECTURER- TS. LAW TENG YI

MEMBERS :

MOHIB BIN AAMIR (TP085584)
ZEBWA WAHEED PARKAR (TP084954)
SMEERAJ KRISHNA (TP082199)

PRESENTATION OUTLINE

- Introduction
- System Overview
- Workload Matrix
- Use Case Diagram
- Class Diagram
- Program Demo
- Limitations
- Conclusion
- Q&A

INTRODUCTION

- CRS is a Java GUI-based system to help institutions manage students who failed components and need recovery support.
- Motivation: reduce manual tracking, improve accuracy, organize recovery workflows.
- Target users: Academic Officer / Course Administrator
- Overall goal: manage recovery planning, eligibility checks, reports, and notifications in one system.

SYSTEM OVERVIEW

Key features/modules in CRS :

- User Management (login, create/update/deactivate users)
- Import & view Students/Courses/Grades
- Eligibility Check & Enrolment (CGPA + failed-course rules) Recovery Plan + Milestones + Progress tracking
- Academic Performance Report + PDF export
- Email notifications (account, enrolment, reports, recovery updates)



OOP Concepts Applied (Quick Summary).

- Encapsulation: private fields + getters/setters for safe access
 - Inheritance: role-based classes (e.g., User → Student/AcademicOfficer/Admin)
- Polymorphism: same action behaves differently depending on role/module usage
- Abstraction: hide complex logic (eligibility, email sending, report generation)
 - Modularity + Packages: separate code into GUI/model/service/util

SYSTEM ARCHITECTURE & PACKAGES

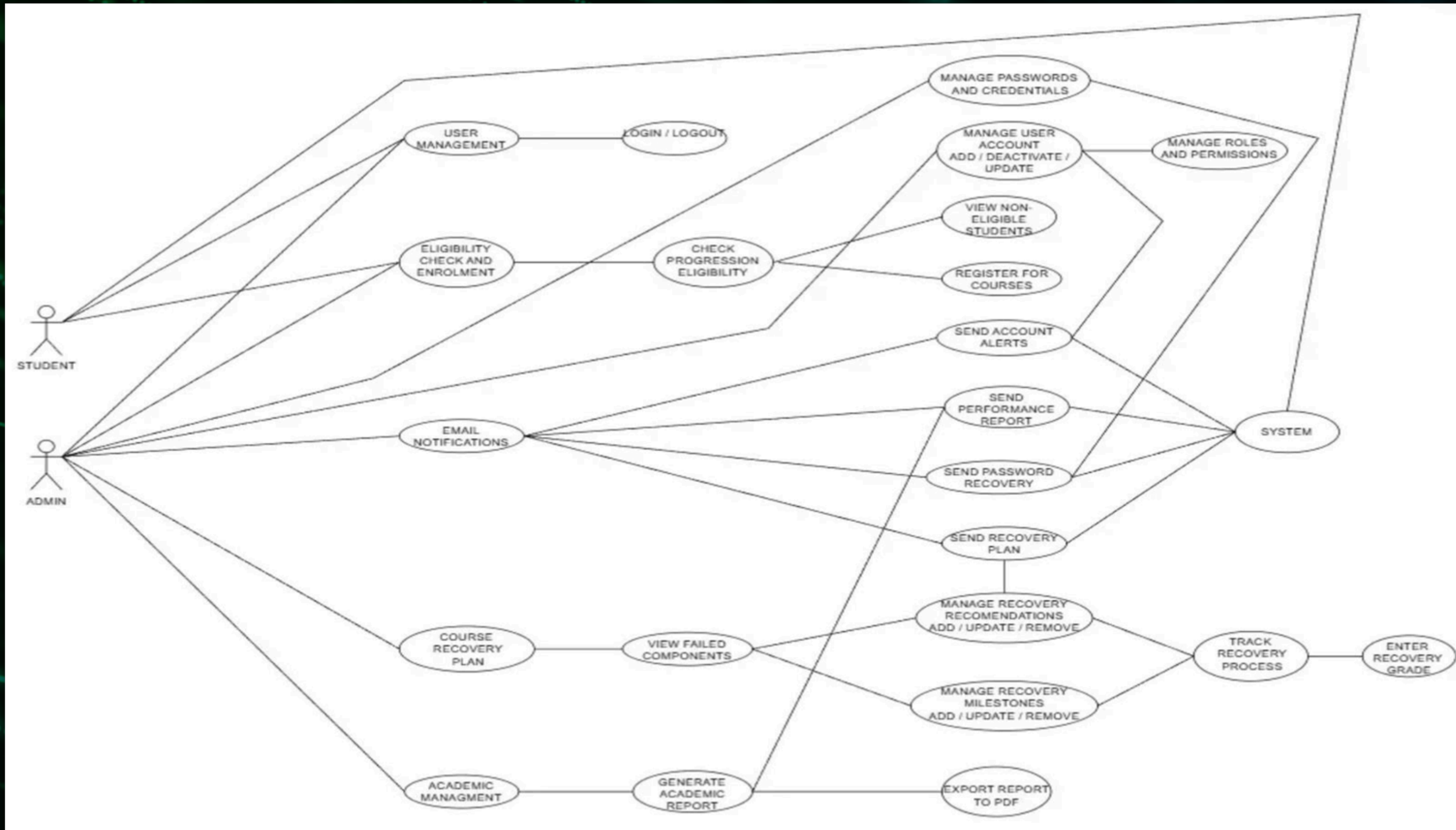
- GUI Layer: Panels/screens (Login, Dashboard, Students, Courses, Eligibility, RecoveryPlan, Reports, UserManagement)
- Model Layer: User, Student, Course, RecoveryPlan, Milestone, etc.
- Service Layer: StudentService, CourseService, UserService, ReportService, RecoveryPlanService, EmailService
- Utility Layer: FileManager, Validation, Password utilities
- Benefit: separation of concerns → easier testing + maintenance



WORKLOAD MATRIX

Names & Task	User Managemen t	Course Recovery Plan	Eligibility Check	Academic Performanc e Reporting	Email	Signature
Mohib Bin Aamir TP085584		100%	100%			Mohib
Smeeraj Krishna TP082199				50%	100%	Smeeraj
Zeba Waheed Parkar TP084954	100%			50%		Zeba

USE CASE DIAGRAM



USE CASE DIAGRAM

Actors: Administrator and Student

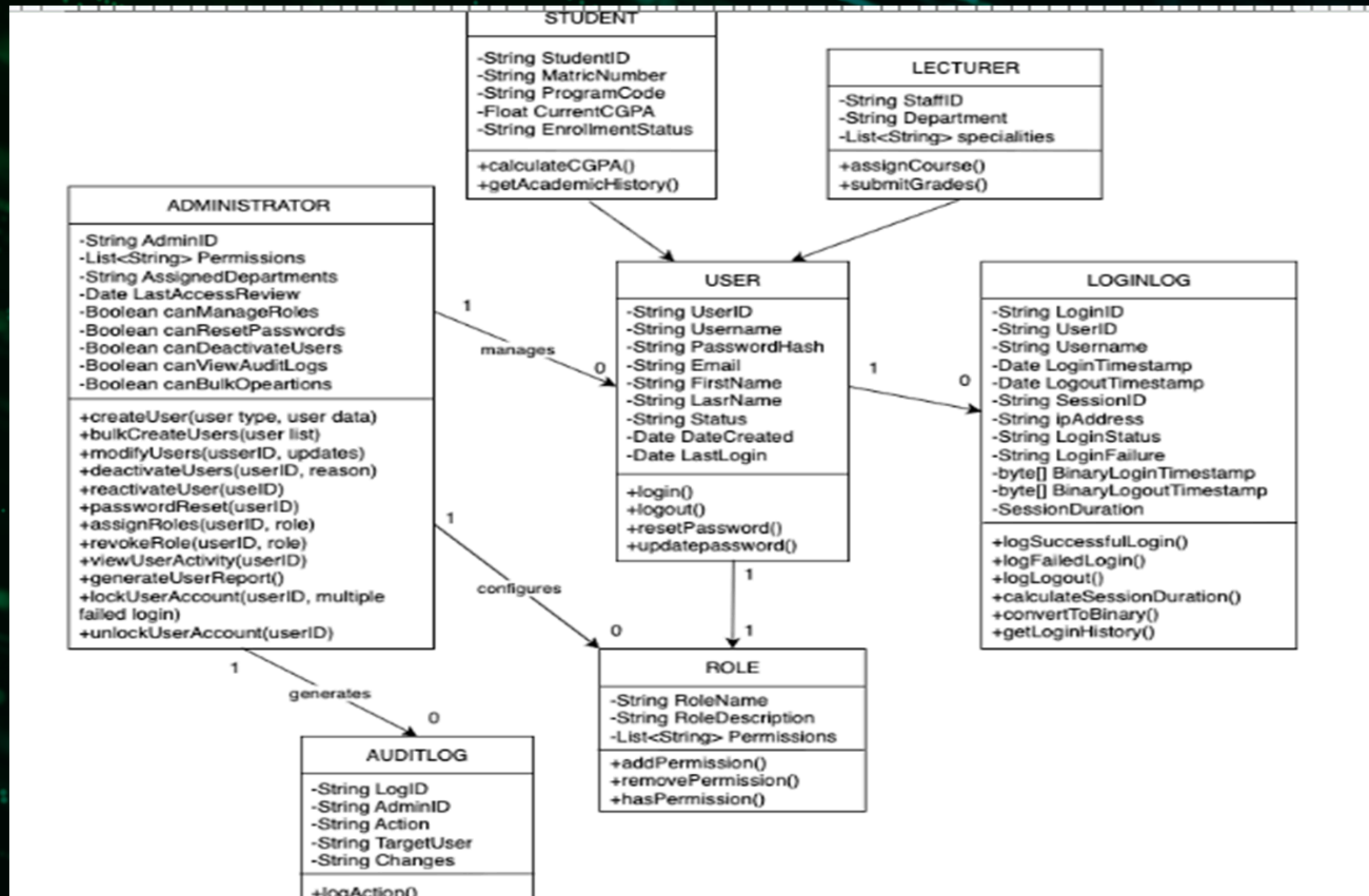
Main Use Cases:

- Login/Logout
- Manage Users (Add/Update/Deactivate/Recover password)
- Import Students/Courses/Grades
- Eligibility Check & Enrolment
- Manage Recovery Plan + Milestones + Progress
- Generate Academic Report + Export PDF
- Send Email Notifications

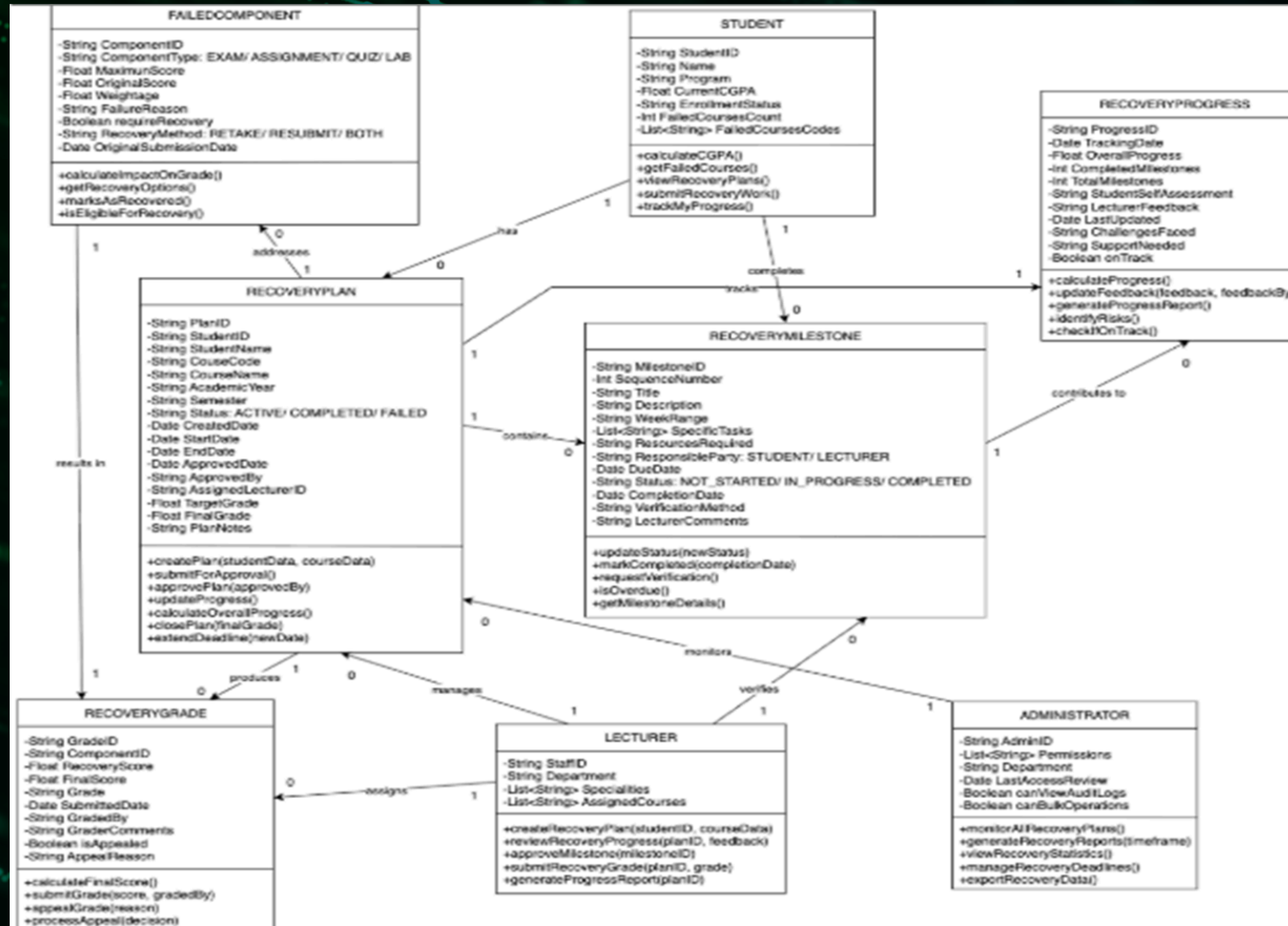
CLASS DIAGRAM

- User Management: User, Student/Admin roles, LoginLog, permissions handling
- Course Recovery: RecoveryPlan + Milestones + progress update workflow
- Eligibility: academic history/grades → eligibility result → enrolment decision
- Email: templates/rules → EmailService → log/status of delivery

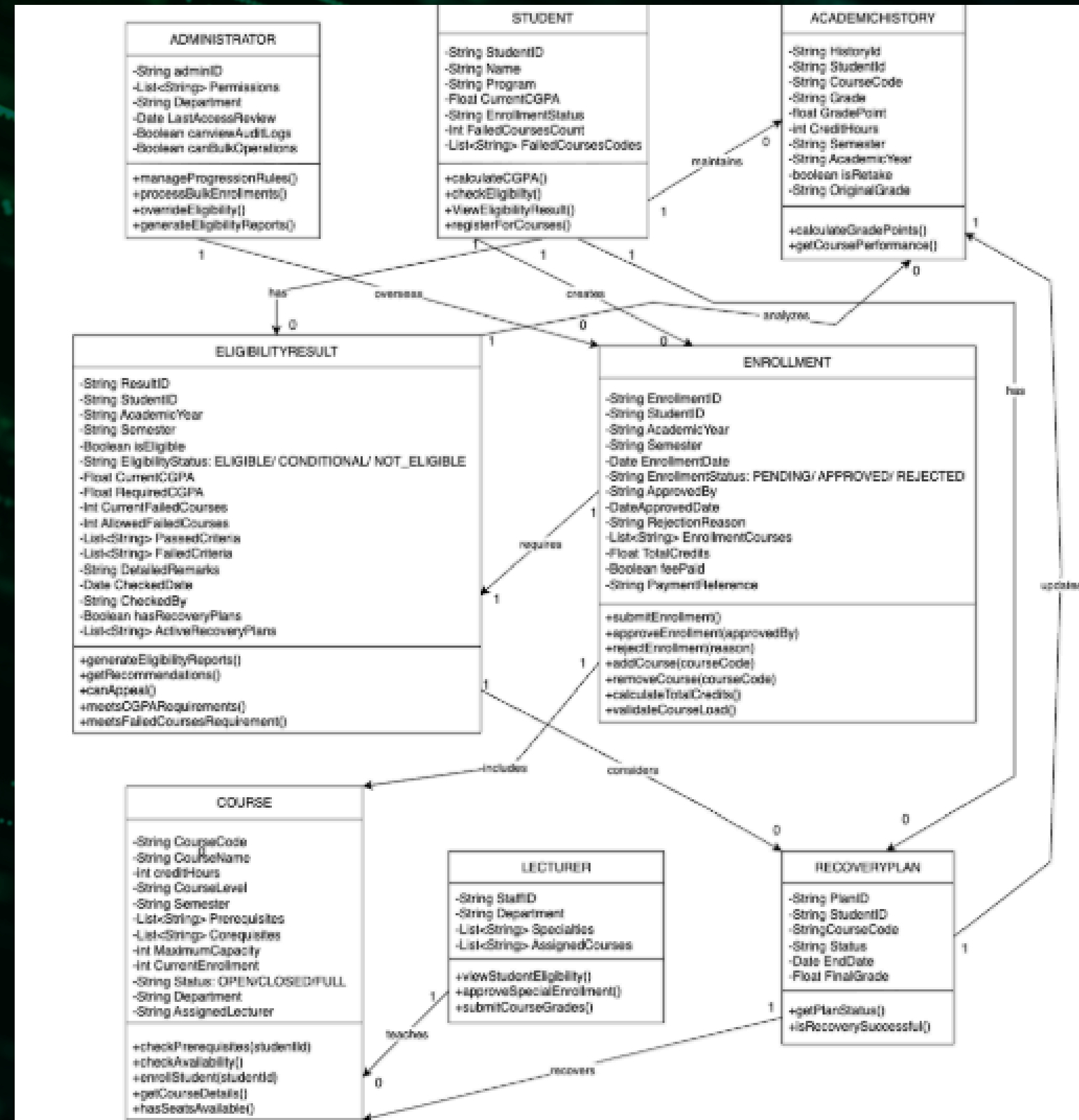
CLASS DIAGRAM– USER MANAGEMENT



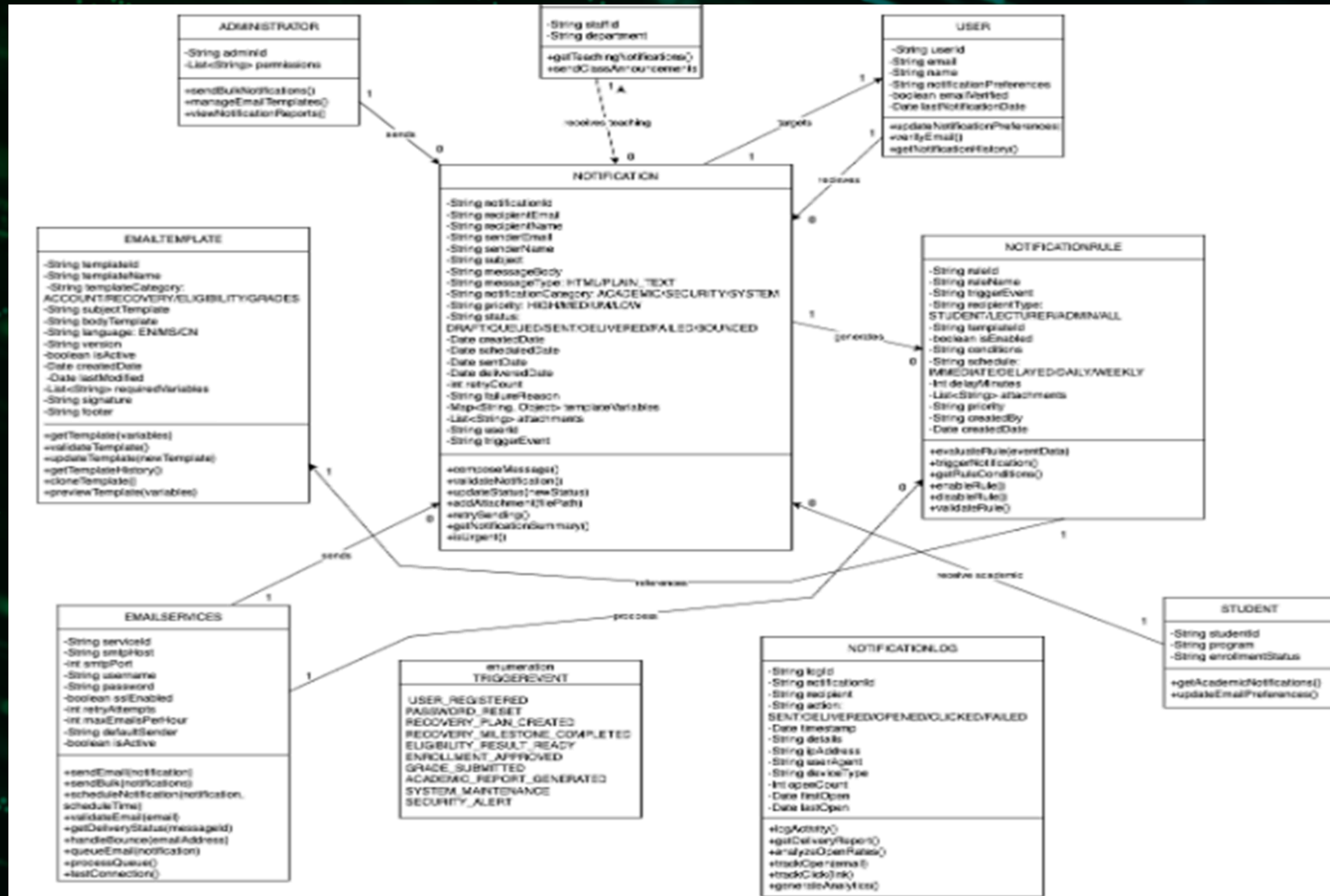
CLASS DIAGRAM- COURSE RECOVERY



CLASS DIAGRAM- ELIGIBILITY



CLASS DIAGRAM-EMAIL





PROGRAM DEMO



LOGIN → DASHBOARD

IMPORT STUDENTS / COURSES / GRADES

RUN ELIGIBILITY CHECK → SHOW ELIGIBLE VS NOT ELIGIBLE
REASON

CREATE RECOVERY PLAN FOR FAILED STUDENT → ADD MILESTONES
→ UPDATE PROGRESS

GENERATE ACADEMIC PERFORMANCE REPORT → EXPORT PDF

SEND EMAIL NOTIFICATION → SHOW SUCCESS CONFIRMATION

PROGRAM DEMO

LIMITATIONS

- FILE-BASED STORAGE (TEXT/BINARY) LIMITS SCALABILITY & CONCURRENCY VS DATABASES
- BASIC SECURITY ONLY (NO MFA, ADVANCED ENCRYPTION/SESSION HANDLING)
- GUI IS FUNCTIONAL BUT MINIMAL (NO ACCESSIBILITY/RESPONSIVE DESIGN)
- FIXED RULES (ELIGIBILITY CRITERIA & REPORT FORMATS NOT EASILY CONFIGURABLE)
- NO INTEGRATION WITH LMS/STUDENT PORTAL DUE TO PROJECT SCOPE

CONCLUSION

CRS ACHIEVED GOALS: USER MANAGEMENT, RECOVERY
PLANNING, ELIGIBILITY CHECKING, REPORTING, EMAIL
AUTOMATION

DEMONSTRATED OOP IN PRACTICE: ENCAPSULATION,
INHERITANCE, POLYMORPHISM, ABSTRACTION, MODULARITY
FUTURE ENHANCEMENTS: DATABASE SUPPORT, STRONGER
SECURITY, IMPROVED UI/UX, FLEXIBLE POLICY RULES



Q & A

THANK YOU

